



9947 - Spectroscopic Exploration of Luminous $z > 9$ Galaxies: Revealing Early Overdensity Peaks with JWST

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Prof. Pascal Oesch (PI) (ESA Member)	University of Geneva, Department of Astronomy
Dr. Christina C Williams (CoI) (US Admin CoI)	University of Arizona
Dr. Rohan Naidu (CoI)	University of Hawaii
Dr. Gabriel Brammer (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Mr. Andrea Weibel (CoI) (ESA Member)	University of Geneva, Department of Astronomy
Prof. Pratika Dayal (CoI) (CSA Member)	Canadian Institute for Theoretical Astrophysics
Dr. Anne Hutter (CoI) (ESA Member)	University of Vienna
Dr. Romain Alexis Meyer (CoI) (ESA Member)	University of Geneva, Department of Astronomy
Dr. Zhiyuan Ji (CoI)	University of Arizona
Dr. Mengyuan Xiao (CoI) (ESA Member)	University of Geneva, Department of Astronomy
Prof. Michael Maseda (CoI)	University of Wisconsin - Madison
Prof. Ivo Labbe (CoI)	Swinburne University of Technology
Prof. Sinclair Manning (CoI)	University of Massachusetts - Amherst
Dr. Callum Witten (CoI) (ESA Member)	University of Geneva, Department of Astronomy

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRSpec				
	1	j0314-pt1-49-full-v1	NIRSpec MultiObject Spectroscopy	(7) PANO_j031404m1759_hzmask
	2	j0314-pt2-49-full-v1	NIRSpec MultiObject Spectroscopy	(7) PANO_j031404m1759_hzmask
	3	j0144-pt1-29-full-v1	NIRSpec MultiObject Spectroscopy	(5) PANO_j014428p1715_hzmask

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
	4	j0218-pt1-36-full-v1	NIRSpec MultiObject Spectroscopy	(6) PANO_j021824m0517_hzmask
	5	j0332-pt1-79-full-v1	NIRSpec MultiObject Spectroscopy	(8) PANO_j033224m2756_hzmask

ABSTRACT

One of the biggest surprises in the first years of JWST has been the discovery of an unexpectedly large number of luminous ($M_{UV} < -20$), and potentially massive ($\log M > 9$) candidates very early in the Universe's history. These findings show a more than 10-fold excess of bright galaxies compared to pre-JWST models, suggesting more efficient star formation, different stellar populations, or other changing physics in the early Universe. However, while photometric redshifts have identified many $z > 9$ candidates, most sources exhibit secondary peaks in their redshift likelihoods. Spectroscopic follow-up is crucial to ruling out low-redshift contaminants such as obscured galaxies or faint AGN -- and to build a library of such spectra. Deep NIRSpec observations have proven very successful to confirm galaxies at $z > 9$, including the luminous GSz14 at $z = 14.18$. Its implied past star-formation history shows that progenitor galaxies as bright as $M_{UV} \sim -19.5$ existed out to $z \sim 18$. Here, we propose to confirm such sources. In particular, we exploit NIRCам pure parallel imaging data, which has the unique power to identify overdense lines-of-sight containing several luminous $z > 9$ candidates at the same time. These fields make NIRSpec multi-object spectroscopy particularly efficient, allowing us to target up to 50 luminous galaxies with $M_{UV} < -19.5$ at $z > 9$, including several promising candidates up to $z \sim 15-17$. Our program has the promise of characterizing the most overdense regions in the early Universe and to reveal the physical properties of a statistical sample of luminous $z > 9$ sources.

OBSERVING DESCRIPTION

The primary goal of this program is to obtain spectroscopic redshift confirmations of bright high-redshift galaxy candidates. We will achieve this with 3.5hr deep NIRSpec/prism observations. No pre-imaging is required, as these sources all have been identified in the NIRCам images of key legacy fields, which were aligned to Gaia DR3.

NIRSpec exposures:

The observing strategy for the MSA consists of 3 nods in 3 slitlet slits, which we repeat 3 times. We are taking a series of 19 groups of integrations and 1 integration per exposure with the NRSIRS2 readout mode, resulting in a total science time of 12605s, or 3.5hrs per mask. This same strategy will be repeated for each of our five masks in the four PANORAMIC fields. Our tests show that we can target up to ~ 50 $z > 9$ galaxies. The MSA mask design will be finalized once the VPA3 angle of JWST is assigned. We have no particular constraints and can safely avoid the micro meteorites.

Background Limited Special Requirement:

One of the requirements of our program is to measure faint continuum spectra, at $\sim 2.0\text{-}4.5\mu\text{m}$. The peak S/N at this wavelength changes by $>16\%$ between the lowest and highest background conditions for one of our pointings, and $>8\%$ for two others. We thus set the Background-Limited Special requirement. This still ensures large enough scheduling windows.

Proposal 9947 - Targets - Spectroscopic Exploration of Luminous $z > 9$ Galaxies: Revealing Early Overdensity Peaks with JWST

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(5)	PANO_j014428p1715_hzmask	RA: 01 44 26.0795 (26.1086646d) Dec: +17 14 31.55 (17.24210d) Equinox: J2000		
	Comments: Description=[]				
	(6)	PANO_j021824m0517_hzmask	RA: 02 18 24.0025 (34.6000104d) Dec: -05 17 21.75 (-5.28938d) Equinox: J2000		
	Comments: Description=[]				
	(7)	PANO_j031404m1759_hzmask	RA: 03 14 2.4145 (48.5100604d) Dec: -17 58 39.07 (-17.97752d) Equinox: J2000		
	Comments: Description=[]				
	(8)	PANO_j033224m2756_hzmask	RA: 03 32 17.4972 (53.0729050d) Dec: -27 58 37.91 (-27.97720d) Equinox: J2000		
	Comments: Description=[]				

Proposal 9947 - Observation 1 - Spectroscopic Exploration of Luminous $z > 9$ Galaxies: Revealing Early Overdensity Peaks with JWST

Observation	Proposal 9947, Observation 1: j0314-pt1-49-full-v1										Thu Apr 23 16:00:56 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(7)	PANO_j031404m1759_hzmask	RA: 03 14 2.4145 (48.5100604d) Dec: -17 58 39.07 (-17.97752d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold
	MSATA	false		After Target ACQ	MSA Center	primary (99 sources)				jwst-nirspec-prism	1.5
Reference Stars											
Confirmation	#	Confirmation Type		Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp		Conf. Total Integrations		Conf. Total Exposure Time	
	1	After Target Acq		NRSIRS2RAPID	5	1		1		87.533	

Proposal 9947 - Observation 1 - Spectroscopic Exploration of Luminous $z > 9$ Galaxies: Revealing Early Overdensity Peaks with JWST

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	48.50658625 Degrees - 17.986183333333 315 Degrees	49.001054236076 72			3	3	4201.6
	2	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	48.50658625 Degrees - 17.986183333333 315 Degrees	49.001054236076 72			3	3	4201.6
	3	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	48.50658625 Degrees - 17.986183333333 315 Degrees	49.001054236076 72			3	3	4201.6
Special Requirements	Background Limited. Background no more than 30th percentile above minimum MSA Planned Aperture PA 49.0000 to 49.0000 Degrees (V3 270.4254303 to 270.4254303)										

Proposal 9947 - Observation 2 - Spectroscopic Exploration of Luminous $z > 9$ Galaxies: Revealing Early Overdensity Peaks with JWST

Observation	Proposal 9947, Observation 2: j0314-pt2-49-full-v1										Thu Apr 23 16:00:56 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(7)	PANO_j031404m1759_hzmask	RA: 03 14 2.4145 (48.5100604d) Dec: -17 58 39.07 (-17.97752d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	MSATA	false	After Target ACQ	MSA Center	primary (99 sources)		jwst-nirspec-prism	1.5			
Reference Stars											
Confirmation	#	Confirmation Type	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time				
	1	After Target Acq	NRSIRS2RAPID	5	1	1	87.533				

Proposal 9947 - Observation 2 - Spectroscopic Exploration of Luminous $z > 9$ Galaxies: Revealing Early Overdensity Peaks with JWST

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	48.509619166666 67 Degrees - 17.958786111111 124 Degrees	49.000175109471 2			3	3	4201.6
	2	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	48.509619166666 67 Degrees - 17.958786111111 124 Degrees	49.000175109471 2			3	3	4201.6
	3	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	48.509619166666 67 Degrees - 17.958786111111 124 Degrees	49.000175109471 2			3	3	4201.6
Special Requirements	Background Limited. Background no more than 30th percentile above minimum MSA Planned Aperture PA 49.0000 to 49.0000 Degrees (V3 270.4254303 to 270.4254303)										

Proposal 9947 - Observation 3 - Spectroscopic Exploration of Luminous $z > 9$ Galaxies: Revealing Early Overdensity Peaks with JWST

Observation	Proposal 9947, Observation 3: j0144-pt1-29-full-v1										Thu Apr 23 16:00:56 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 3:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(5)	PANO_j014428p1715_hzmask	RA: 01 44 26.0795 (26.1086646d)								
			Dec: +17 14 31.55 (17.24210d)								
	Equinox: J2000										
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	MSATA	false	After Target ACQ	MSA Center	primary (73 sources)		jwst-nirspec-prism	1.5			
Reference Stars											
Confirmation	#	Confirmation Type	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time				
	1	After Target Acq	NRSIRS2RAPID	5	1	1	87.533				

Proposal 9947 - Observation 3 - Spectroscopic Exploration of Luminous $z > 9$ Galaxies: Revealing Early Overdensity Peaks with JWST

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	26.111264541666 667 Degrees 17.227278055555 555 Degrees	29.000739288137 407			3	3	4201.6
	2	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	26.111264541666 667 Degrees 17.227278055555 555 Degrees	29.000739288137 407			3	3	4201.6
	3	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	26.111264541666 667 Degrees 17.227278055555 555 Degrees	29.000739288137 407			3	3	4201.6
Special Requirements	Background Limited. Background no more than 30th percentile above minimum MSA Planned Aperture PA 29.0000 to 29.0000 Degrees (V3 250.42543030000002 to 250.42543030000002)										

Proposal 9947 - Observation 4 - Spectroscopic Exploration of Luminous $z > 9$ Galaxies: Revealing Early Overdensity Peaks with JWST

Observation	Proposal 9947, Observation 4: j0218-pt1-36-full-v1										Thu Apr 23 16:00:56 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(6)	PANO_j021824m0517_hzmas k	RA: 02 18 24.0025 (34.6000104d) Dec: -05 17 21.75 (-5.28938d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	MSATA	false	After Target ACQ	MSA Center	primary (31 sources)		jwst-nirspec-prism	1.5			
Reference Stars											
Confirmation	#	Confirmation Type	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time				
	1	After Target Acq	NRSIRS2RAPID	5	1	1	87.533				

Proposal 9947 - Observation 4 - Spectroscopic Exploration of Luminous $z > 9$ Galaxies: Revealing Early Overdensity Peaks with JWST

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	34.604951791666 664 Degrees - 5.268547222221 96 Degrees	35.999583794832 37			3	3	4201.6
	2	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	34.604951791666 664 Degrees - 5.268547222221 96 Degrees	35.999583794832 37			3	3	4201.6
	3	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	34.604951791666 664 Degrees - 5.268547222221 96 Degrees	35.999583794832 37			3	3	4201.6
Special Requirements	Background Limited. Background no more than 30th percentile above minimum MSA Planned Aperture PA 36.0000 to 36.0000 Degrees (V3 257.4254303 to 257.4254303)										

Proposal 9947 - Observation 5 - Spectroscopic Exploration of Luminous $z > 9$ Galaxies: Revealing Early Overdensity Peaks with JWST

Observation	Proposal 9947, Observation 5: j0332-pt1-79-full-v1										Thu Apr 23 16:00:56 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(8)	PANO_j033224m2756_hzmas k	RA: 03 32 17.4972 (53.0729050d) Dec: -27 58 37.91 (-27.97720d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	MSATA	false	After Target ACQ	MSA Center	primary (73 sources)		jwst-nirspec-prism	1.5			
Reference Stars											
Confirmation	#	Confirmation Type	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time				
	1	After Target Acq	NRSIRS2RAPID	5	1	1	87.533				

Proposal 9947 - Observation 5 - Spectroscopic Exploration of Luminous $z > 9$ Galaxies: Revealing Early Overdensity Peaks with JWST

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	53.056301666666 66 Degrees - 27.991920833333 324 Degrees	79.007748047867 35			3	3	4201.6
	2	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	53.056301666666 66 Degrees - 27.991920833333 324 Degrees	79.007748047867 35			3	3	4201.6
	3	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	53.056301666666 66 Degrees - 27.991920833333 324 Degrees	79.007748047867 35			3	3	4201.6
Special Requirements	MSA Planned Aperture PA 79.0000 to 79.0000 Degrees (V3 300.4254303 to 300.4254303)										